



Framework for comparison of data visualization tools and an illustration using Power BI, QlikSense and Google Looker

Abstract:

Global data visualization market is growing at a rate of ~10% and is expected to reach \$19.20 B by 2027 ^[5]. Given its market size, wide range of data visualization platforms and products are available in the market. The platforms offer dashboards, analytics, scorecards, and business intelligence (BI) solutions to identify various data trends and patterns. In this paper, we have created a framework with set of parameters, which would help organizations to evaluate the data visualization platform or product. We have discussed and compared application of parameters over three business intelligence and visualization platforms – Power BI, Google Looker and QlikSense.

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Introduction

Hans Rosling, a leading public health educator, statistician and epidemiologist, is well known for his TED talks^[1]. Rosling's TED talk centered around the evolution of the human population in the last 200 years is viewed by many on YouTube. Rosling's leveraged power of data visualization to tell a powerful story around evolution of human health and wealth in four minutes. The data visualization nicely combines different attributes and dimensions such as Life Expectancy, Wealth, Population Growth, Geography and time evolution.

The above is only one example to illustrate the importance of data visualization. Over a period, the interest in data visualization tools to interpret, analyze and generate insights from data have considerably increased. The plethora of tools available now, also poses challenges in choosing the right set of tools and having a set of parameters to choose the visualization tools for the enterprise. This white paper provides a set of parameters for selection of data visualization tools and provide an illustrative comparison using three popular data visualization tools – Power BI, Google Looker and QlikSense.

Overview of Power BI

Microsoft provides below types of Power BI product variants -

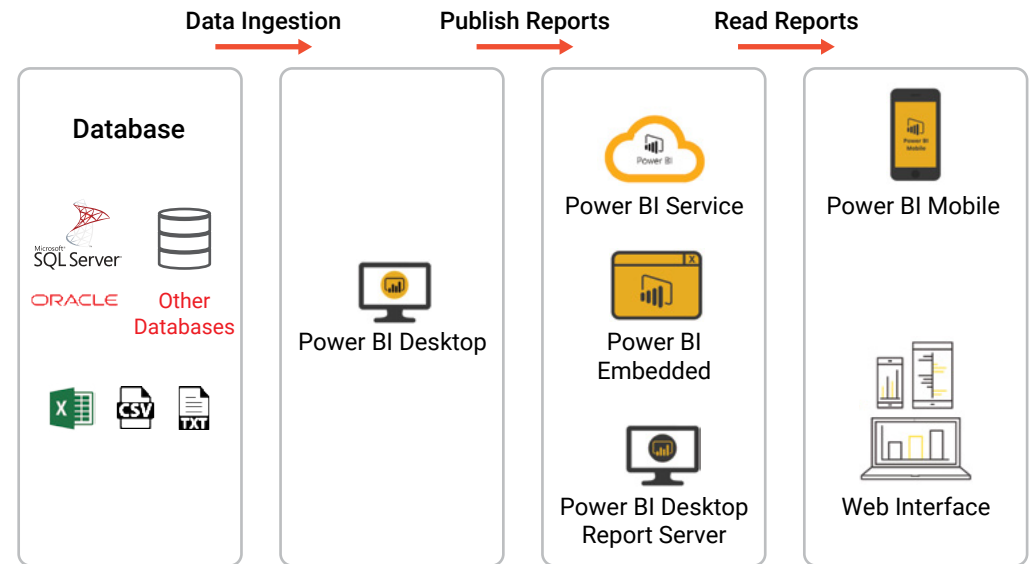
- Power BI Desktop: A desktop application to create data models and reports
- Power BI Service (SaaS i.e., Software as a Service – cloud-based report sharing platform)
- Power BI Mobile (for android and iOS devices)
- Power BI report Server: On Premises report server to share reports, dashboards within organization
- Power BI Embedded: Allows developers to embed Power BI dashboards and reports into other applications, within organization.

Power BI can be deployed both on-premises and on-cloud. It can import data from multiple data sources such as local databases, cloud database, big data sources, simple Excel files, and other hybrid sources. Power BI provides close integration with Microsoft Office 365 products.



Power BI consists of below key components:

- **Power Query:** Power Query Editor is used to connect to one or many data sources, shape and transform the data.
- **Power Pivot:** Used for in memory data modeling and analysis
- **Power View:** To create a variety of visualizations, from tables and matrices to pie, bar, and bubble charts and sets of multiple charts
- **Power Map:** Used for visualization of 3D geo-spatial data
- **Power Q&A:** Q&A feature in Power BI lets you explore your data in your own words by asking questions in natural language. For example, “what were total sales last year?”



Above diagram shows different Power BI products and their functions. We selected Power BI for comparison as it's the leader in BI tools market.



Overview of Qlik cloud & Qlik Sense

Qlik provides Qlik cloud, a comprehensive cloud-based solution that provides the following:

- Data Management services
- Data Integration, Data Transformation
- Analytics & visualization
- Alerting & Notifications
- Embedded analytics

In the scope of this white paper, Qlik Sense is chosen as it is the latest analytics and visualization solution from Qlik. Qlik Sense provides, self-service visualization, interactive dashboards, conversational analytics, NLP and catalogues.

Qlik Sense consists of Qlik-managed cloud-based solutions, viz. Qlik Sense Enterprise SaaS and Qlik Sense Business; and a customer-managed solution, viz. Qlik Sense Enterprise Client-Managed.

Qlik Sense uses in-memory, columnar storage and an associative engine that is meant for high performance analytics use cases. Qlikview is the classic visualization product from Qlik and of late, more market adoption is seen for the QlikSense solution. QlikSense Enterprise can host both QlikSense applications and QlikView applications, collectively known as Qlik apps.



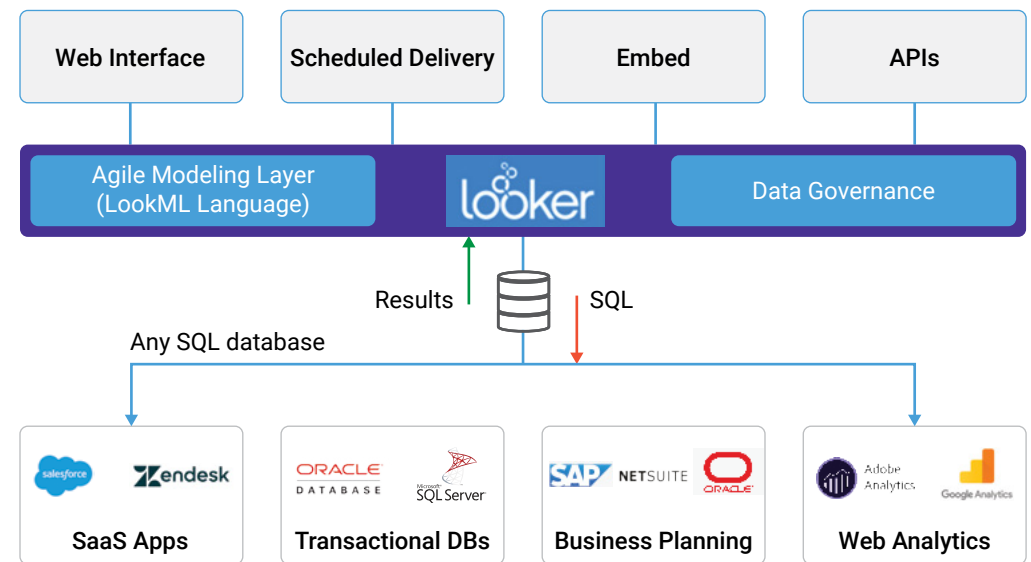
Overview of Google Looker

Looker is a cloud based modern business intelligence and data application platform (acquired by Google in 2020). The architecture of looker is designed to provide scalability and performance by running workloads on your choice of clouds and databases. Looker has a sophisticated data modeler that intelligently scans data and infers relationships between tables in schema to build a basic model. The data modeler enables analysts to connect directly to raw data without any intermediate tools. With Looker, it is easy to automate processes and create insights with data regardless of its source. Looker supports around 60 database dialects. Looker allows integration with 3rd-party tools thorough the use of inbuilt APIs. Looker platform has below features:

- **LookML:** Looker is based on a SQL-based modeling language called LookML. Analysts can use LookML to centrally define and manage definitions and business rules in single Git and version-controlled data model.
- **Embedded Analytics:** Looker's embedded analytics feature enables data analysts, developers, and product managers to rapidly create custom applications and embed analytics in existing applications, websites, and portals.
- **Data visualizations, dashboards, and Looks:** Looker provides interactive dashboards and Looks (component dashboards) that enable real-time data experiences governed by trusted and unified data model.
- **Looker API:** Looker provides RESTful APIs for platform customization and data query. APIs can be used to create custom applications and workflows or scripts that automate tasks such as Schedule Repots, Embed Insights to website, Provision New Looker Accounts etc.

- **Looker Blocks and Actions:** These are prebuilt pieces of code that can accelerate the development cycle. Looker Blocks include optimized SQL patterns, data source connections, customized analytics, and interactive visualizations.

Below diagram shows functional architecture of Google Looker ^[4]



In this whitepaper, we selected Looker for comparison as this is one of the fastest growing BI tools and has unique features.



Parameters used for Comparison

- **Data connectors:** The flexibility in the architecture and design of the solution to connect to a variety of data sources such as the following is considered here.
 - » Ability to connect to on-premise and on-cloud data sources
 - » Big data sources,
 - » Relational databases,
 - » No SQL databases,
 - » Cloud data stores and cloud Datawarehouse such as Redshift
 - » Connectors to 3rd party social media services such as Facebook, Google analytics
 - » Different File formats such as excel worksheets, CSV etc.
- **Data Transformation:** The facilities provided by the BI platform to extract, load, transform data is considered here. BI platforms need to provide mechanism to load certain subset/certain columns of the data, perform different operations such as joins, aggregation and create intermediate data stores for data visualizations.
- **Dashboards, Views and Charts:** Availability of tools, development environments to create different charts, views, filters, drop downs etc. are considered here. Basic chart types such as Line charts, Pie charts, Scatter plots, Bubble Charts, Heat Maps, Tree maps to advanced charts such as word clouds, stacked charts and geospatial plots are available from the data visualization tools now.
- **Data Analytics:** The ability of the BI platform to provide in-built analytics features such as supervised, semi-supervised or unsupervised Machine Learning Models, Natural Language processing (NLP)/text-based analytics, neural networks based prediction or the ability to integrate with any third-party analytics frameworks such as python, tensorflow, R is considered here. In this case, the outputs from the analytics models can be used to create the contents or inputs for a report.
- **Reports publishing & integration:** Reports that are created need to be published for consumption of business users and many times must be integrated with other applications/websites as well. Also, the reports shall be available on different form factors, screen resolutions and devices for reaching to multiple users. Hence, the availability of reports on mobile, tablet and desktop/laptop form factors are being factored into this.



- **Deployment:** The ease of deploying, installing, and carrying out the routine administrative tasks is considered here.
- **Product variants & pricing:** BI platforms provide different subscription mechanisms and licensing options and product variants for enterprise usage. It helps the enterprises to choose the right product for their usage and optimize their spend.
- **Market Place & Community support:** A thriving developer community where the business reports developers can seek clarifications and help and a marketplace to promote BI solutions is critical for the success.
- **Multi language support:** The ability of the BI tool to use either operating system locale or the browser language preference and display the content on languages other than english such as Japanese, German, or Chinese and critical for business users is considered here.
- **Multi-Platform support:** Availability of mobile application for rendering/ consuming reports and the ability to view the reports in different form factors such as laptops, desktops, tablets is considered here.
- **Real Time reporting/Notifications:** Real time and near real time reporting capability is critical for business users in many segments. Also, the ability to provide notifications on different channels for collaboration such as Slack, Email is critical for adopting the BI tool within an enterprise.
- **Security:** Role based access controls and ability to restrict data access for certain rows, based on specific filters and roles is critical to ensure that certain reports are available only to authorized set of users within an enterprise. In addition, user administration and user authentication, and auditing platform access is an essential parameter considered by an enterprise for BI tool adoption.
- **Performance on large data volume:** BI tools shall provide the capability to work with large volumes of data and generate or update reports with performance expected by the business user.
- **Version Control:** System of records for changes in file with respect to data modeling and transformation (Through Git and BitBucket)
- **Semantic Layer Building:** Exposing the transformed data (for model building) to multiple platforms (external platforms)
- **Support for scripting languages for loading, modelling of data:** Backend scripting language for ease of data modeling, visualization, and customization



Detailed Comparison

We compared the BI tools and rated them on the basis assessment criteria. The comparison is given in below table.

CRITERIA	Power BI ^[3]	Qlik Sense	Google Looker ^[2]
Data connectors	- Various data sources are supported such as sql, Azure, excel, JSON, Hadoop file, ODBC, Python and R script, spark etc - Data connectors extend the ability to access data from 3rd-party applications, web services, offline or online data sources.	- Connect data from hundreds of data sources – from apps, databases, cloud services and more. - QlikSense allows to unlock siloed data and delivers comprehensive views of business to generate new insights and data-driven actions.	- Looker connects with Redshift, Snowflake, BigQuery, and 50+ supported SQL dialects. - Can link multiple databases, avoids database lock-in, and maintain multi cloud data environments.
Dashboard/ Views	- Dashboard is Key feature of Power BI and has a clean UI to publish the report. - A wide selection of visualization types to choose from, including Donut Chart, Funnel Chart, and Ribbon Chart.	Wide range of visualization options: Pie Chart, Histogram, Treemap, Bar chart, Pivot table, Combo chart, Scatter plot, Box plot, and Distribution plot. Also, creates custom widgets with multiple dimensions.	Can create dashboards from Looks and Tiles. Options for navigating dashboards, exploring, and drill down on data points, filters, dashboard LookML, pinning dashboards, and also allows downloading data from dashboards.



Performance/ Volume of data	• Enable the Large dataset storage format option to use datasets in Power BI Premium that are larger than the default limit. The dataset size limit (for premium version) is comparable to Azure Analysis Services	• Data engine allows exploration of all data in a complete suite of visualizations. Also allows compression of large data sets, and highlights outliers	• Caching of data can be controlled to optimize performance and latencies for large data sets • The in-memory analysis service for Big Query allows users to analyze large and complex datasets interactively in Looker
Report Publish and Integration	• Reports can be published on cloud or on-premises server. • Reports and dashboards can be embedded to website portals (internal, public)	• Visualizations and reports can be shared as image, link, or embedded HTML • Visualizations can be downloaded in cloud hub apps as images or PDF files	• Looks, Explores and dashboards can be published on web (internal, public) through URLs • Looker also supports APIs
Transformation/ Data modelling	• Allows creation of relationship to create a logical connection between different data sources. • Allows graphical interaction with the data model/relationships.	• Allows creation of the relationships of data tables and data models.	• Models and transformations can be done using LookML • Looker intelligently scans the data and infers relationships between tables in the schema to build a basic model. This reduces data modeling efforts.



Data Analytics	• Connects to R and Python. • Quick Insights, Key influencers, and Q&A – AI based features were added on to Power BI. • Has analytics capabilities like clustering, forecasting and decision-making trees.	• Offers complete range of AI capabilities built into the platform level. • Suggests insights and new connections when data is explored. • Also, supports Python/ R. to perform analytics features like regression, predictive analysis, and connection to third-party apps	• Looker has platform-exclusive analytics function and blocks. The blocks are pre-built and customizable to the user's needs. • Supports Python/R.
Multi-platform	• Power BI has a mobile app for viewing • Power BI cloud is used for uploading and sharing reports.	• Qlik has a native mobile app. The app has interactive offline and online exploration. Integration with alert mechanism to push notifications. • QlikSense can be deployed cloud (SaaS) and/or on-premises	• Looker has a mobile app. • Looker is a cloud-based business intelligence tool.
Security/ Administration	• Power BI has Row Level Security features for data restriction to users.	• Qlik Sense has attribute-based security rules engine to evaluate access grant for users.	• Looker offers row, column security and localization options. Administrators can set permissions by user or group, and can restrict data access from database level to row/ column level



Pricing	• Power BI pro : \$13.70 per user/ month ^[6] • Power BI Premium : \$27.50 per user/ month ^[6] • Capacity license for organization : \$6,858.10 capacity/month ^[6]	• Its subscription starts from \$ 30 per user per month ^[7]	• Approx. \$3,000 per month (combination of developers and viewers licenses for organization)
Product Variants	• PowerBI Desktop (Reports can be shared with others directly or by publishing to the Power BI service.) • PowerBI Service (Cloud based service or SaaS) • Power BI Report Server	• QlikSense Desktop • Qlik Sense Cloud • Qlik Sense Server	• Looker Cloud



Assessment Criteria	PowerBI	QlikSense	Looker
Data connectors	High	High	Medium
Data Transformation	High	High	High
Dashboard and Views	High	High	Medium
Data Analytics	Medium	High	Low
Reports publishing & integration	High	High	High
Security and Ease of Administration	High	High	High
Ease of Deployment	High	High	High
Support for Product variants	High	High	High
Marketplace & Community support	High	Medium	Medium
Multi language support	High	High	High
Multi-platform Support	High	High	High
Real Time reporting/Notifications	Medium	High	High
Performance on large Volume of data	Medium	High	High
Version Control	Low	Low	High
Semantic Layer building	Low	Low	High
Support for scripting languages for loading, modelling of data	Medium	Medium	High



References

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4. [Google Looker Architectural Diagram](#)
5. [Data Visualization market size](#)
6. [Power BI license cost](#)
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About Sasken

Sasken is a specialist in Product Engineering and Digital Transformation providing concept-to market, chip-to-cognition R&D services to global leaders in Semiconductor, Automotive, Industrials, Consumer Electronics, Enterprise Devices, SatCom, Telecom, and Transportation industries. For over 32 years and with multiple patents, Sasken has transformed the businesses of 100+ Fortune 500 companies, powering more than a billion devices through its services and IP.

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